

Ben Abraham

benabraham.vercel.app | ben.harry.abraham@gmail.com | linkedin.com/in/drbenabraham

SUMMARY

Researcher and full stack developer with Department of Defense and National Science Foundation funded research spanning quantum networking, machine learning, and IoT security. Combines hands on engineering across machine learning using Python, production web and mobile applications, and a self founded software studio, alongside real world security and intelligence experience. Builds practical, deployable systems while bridging research, software engineering, and security.

EDUCATION

Kean University <i>Master of Science in Computer Science</i>	Union, NJ May 2027
Kean University <i>Bachelor of Science in Information Technology (Minor: Cyber Security)</i>	Union, NJ May 2025

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS
ML & Data: PyTorch, TensorFlow, NumPy, pandas, logistic regression, deep learning, statistical modeling, data cleaning & pipeline design
Research: Quantum network simulation, entanglement fidelity analysis, radio fingerprinting, IoT security
Security & Networking: Penetration testing & CTFs, intelligence analysis, license plate recognition & surveillance systems, RFID, Cisco networking, Wi-Fi 6
Web & Tools: ReactJS, Git, Arduino, 3D modeling & printing, Web hosting, E-Commerce

EXPERIENCE

Graduate Research Assistant <i>Kean University, Department of Psychology (TEEN Lab)</i>	Sept 2026 Union, NJ
- Serve as project manager for a developmental neuroscience study on how VR social stressors affect adolescent physiological and behavioral responses, coordinating multimodal data collection (EEG, actigraphy, salivary hormones, behavioral data) on a fixed timeline. - Support data cleaning, integration, and analysis in Python toward publication, and complete required human subjects and IRB training.	
Graduate Research Assistant <i>Kean University, College of Business and Public Management</i>	Dec 2025 Union, NJ
Designed and executed data cleaning and analysis pipelines in Python for Survey of Income and Program Participation (SIPP) datasets, spanning raw data acquisition and preprocessing across the 1996 to 2008 panels.	
Quantum Researcher <i>Department of Defense, National Science Foundation, Florida International University</i>	Jan 2025 Miami, FL
Designed and implemented quantum network simulations scaling to 1,200 nodes to evaluate how imperfect Bell State Measurements (BSMs) and classical communication delays affect end to end entanglement fidelity and throughput.	
Co-Entrepreneurial Lead <i>National Science Foundation I-Corps Hub (Northeast Region), Team SAGEM</i>	Jan 2025 Northeast Region
- Completed the National Science Foundation I-Corps program as Co-Entrepreneurial Lead on team SAGEM, driving customer discovery, market validation, and commercialization research for a cognitive health product. - Conducted and analyzed 30+ customer interviews to surface market trends, then prepared and delivered cohort presentations to inform the product and business plan.	
Real Time Crime Center Data Analyst <i>New Jersey State Police</i>	Sep 2024 Trenton, NJ
- Monitored and analyzed 9+ real time criminal intelligence feeds, license plate recognition systems, and surveillance databases to support active investigations and field operations. - Produced intelligence reports, data visualizations, and crime pattern analyses to inform law enforcement decision making. - Facilitated information sharing across local, state, and federal partners, strengthening situational awareness and response times.	

Socio-Economic Research Fellowship

Kean University, National Science Foundation

Jan 2024

Union, NJ

- Developed and implemented machine learning models in Python (logistic regression and additional statistical techniques) to predict contingent worker classification as part of a National Science Foundation funded research team.

CAHSI Cyber Security Researcher

University of Texas, Kean University, National Science Foundation

Feb 2024

Union, NJ

- Conducted mentored cybersecurity research through the CAHSI REU program; authored a research plan, tracked progress, and presented findings via a formal research poster.

End User Field Services Technician

NJ Transit (third largest public transit system in the U.S.)

Jan 2024

Maplewood, NJ

- Provided on site technical support across NJ Transit, troubleshooting hardware, software, and network connectivity issues to minimize downtime.
- Configured, maintained, and deployed workstations; supported ticket resolution, system updates, and user training to ensure reliable service delivery.

IoT Research Assistant

Department of Defense, National Science Foundation, Florida International University

May 2023

Miami, FL

- Contributed to development of a deep learning based radio fingerprinting system for IoT device authentication under National Science Foundation and Department of Defense funding.

Cisco Network Intern

Cisco Inc.

Aug 2022

New York, NY

- Collaborated with Cisco and Round Rock engineers to deploy 16 Meraki MR76 and MR86 Wi-Fi 6 access points across VIP and outdoor zones for the Global Citizen Festival in NYC, gaining hands on experience in large scale event networking and wireless infrastructure optimization.

PROJECTS

Websites and Mobile Applications | *ReactJS, TypeScript, HTML, CSS, SQL*

- **Founder, Dynamic.IO:** Founded and operate dynamicio.net, delivering a large portfolio of client projects across mobile applications and websites, overseeing development, deployment, and client collaboration end to end.
- **Community Cyber Security Tools:** Designed and deployed multiple free, community focused cybersecurity web apps, including metadataanalyzerv2.vercel.app, passwordgenanalyser.vercel.app, and genhash.vercel.app.
- **E-Commerce Platforms:** Built and launched revenue generating e-commerce sites (saminusa.com, focalstands.com) on a Shopify backend, implementing product management, secure payments, order fulfillment, and analytics.
- **Web & Mobile Games:** Created a portfolio of web and mobile games (BananaCPM, Flying Birdie, Tic Tac Toe, Rock Paper Scissors), currently being unified under a single gaming site.
- **Personal Portfolio:** Designed and maintain a personal portfolio site showcasing projects, skills, and experience.

COMPETITIONS

- GMIS National Conference CTF (2024), top 25% nationally
- NJIT National CTF (2023 & 2025), top 20% nationally
- Kean University Hackathon (2022), 2nd place; built an RFID door lock on an Arduino Uno (C++) with custom 3D printed parts
- TryHackMe (2021), ranked in the top 2% globally

TRAINING

- TryHackMe, Cyber Defense Learning Path (Threat & Vulnerability Management, Incident Response)
- TryHackMe, Junior Penetration Tester Learning Path

AFFILIATIONS

- IEEE, Member
- ACM, Member (Kean University chapter)

LEADERSHIP

Police Explorers

- Promoted to Chief; Led drills, trained and mentored junior Explorers, and directed squads as Chief
- Promoted to Sergeant
- Contributed hundreds of hours of community service

SkillsUSA

- Served as a judge for the Tech Apps competition (Thomas Edison High School, NYC)